

Experiment – 04 Study and Verification of Standard Network Topologies

Lab Objective:

To understand and verify the structure and functioning of various standard network topologies, specifically **Star**, **Bus**, and **Ring**. This includes learning how these topologies are physically and logically implemented and how they behave in a network environment.

Prerequisites:

- Basic understanding of computer networks
- Knowledge of networking devices (like switches, hubs, NICs)
- Familiarity with IP addressing and network configuration

Network Topologies – Overview:

A **network topology** refers to the arrangement of different elements (links, nodes, etc.) in a computer network. It can be **physical** (actual layout of cables and devices) or **logical** (the way data flows within the network).

The common types of network topologies include:

1. Star Topology:

Structure:

- All computers (nodes) are connected to a central device (hub or switch).

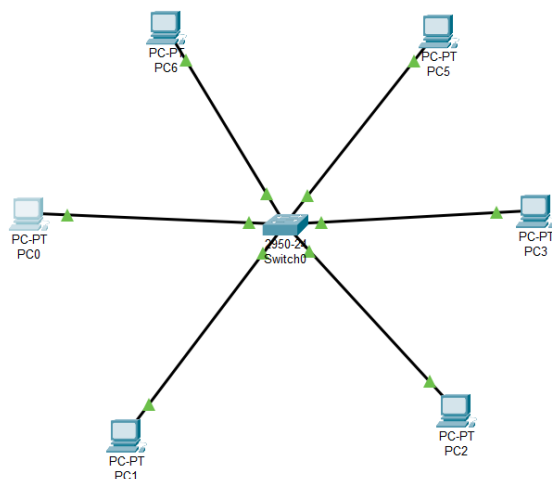
- Each device has a dedicated point-to-point link to the hub/switch.

Advantages:

- Easy to install and manage
- If one cable fails, only that specific node is affected
- Centralized management

Disadvantages:

- If the central hub/switch fails, the whole network goes down
- Requires more cable than bus or ring



2. Bus Topology:

Structure:

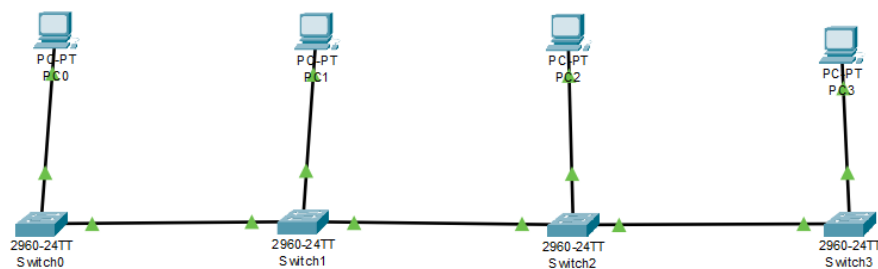
- All computers are connected to a single central cable (called the backbone).
- Data is sent in both directions and received by all devices, but only the intended recipient accepts and processes it.

Advantages:

- Easy to implement and extend
- Requires less cable compared to star topology

Disadvantages:

- Difficult to troubleshoot
- A failure in the backbone affects the entire network
- Data collisions are common in large networks



3. Ring Topology:

Structure:

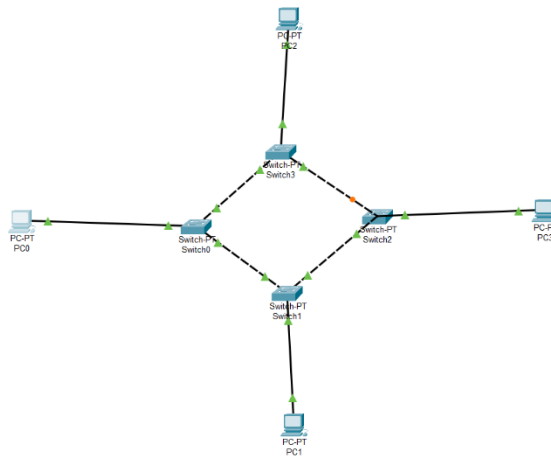
- Each computer is connected to two other computers, forming a circular (ring-like) network.
- Data travels in one direction (unidirectional) or both directions (bidirectional in dual-ring).

Advantages:

- Equal access for all nodes
- Good performance under heavy load

Disadvantages:

- A single node or cable failure can disrupt the entire network (unless a dual-ring or fault-tolerant system is used)
- Difficult to reconfigure and troubleshoot



Outcome:

By the end of this experiment, students will be able to:

- Identify and set up different network topologies
- Understand their structure and performance characteristics
- Observe how data flows through each topology
- Configure network settings and ensure inter-device communication

Steps for Verifying Ring Topology:

Step 1: Gather Required Hardware

- Computers or laptops with **Network Interface Cards (NICs)**
- Network cables (Ethernet)
- Switches or connectors if needed

Step 2: Physical Setup

- Connect each computer to the next in a circular manner using Ethernet cables.
- Ensure that the **last computer is connected back to the first** to complete the ring.
- Example: PC1 → PC2 → PC3 → PC4 → PC1

Step 3: Network Configuration

- Assign IP addresses to all computers within the same network range.
 - Example:
 - PC1: 192.168.1.1
 - PC2: 192.168.1.2
 - PC3: 192.168.1.3
 - PC4: 192.168.1.4
- Set the same **subnet mask** (e.g., 255.255.255.0)
- Ensure all computers can **ping** each other successfully.

Step 4: Verification

- Use the ping command to test connectivity between the computers.
- Observe the data flow and check if the communication is successful.
- Simulate node failure (e.g., disconnect one PC) and observe the impact on the network.

Conclusion:

Through this experiment, we gained a hands-on understanding of **how standard topologies work**, how to **set them up**, and their

practical advantages and disadvantages. Among all, **Star** is the most widely used in modern networks due to its simplicity and reliability, while **Bus and Ring** are mostly found in legacy systems or academic demonstrations.